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► To cite this version:

Bonnaire Céline, Beck François, Spilka Stanislas. Physical and psychosocial factors associated with psychostimulant use in a nationally representative sample of French adolescents: Specificities of cocaine, amphetamine, and ecstasy use. *Addictive Behaviors*, 2019, 92, pp.208 - 224. 10.1016/j.addbeh.2018.11.046 . hal-03485862

HAL Id: hal-03485862

<https://hal.science/hal-03485862v1>

Submitted on 20 Dec 2021

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Physical and psychosocial factors associated with psychostimulant use in a nationally representative sample of French adolescents: specificities of cocaine, amphetamine, and ecstasy use

Bonnaire Céline^{a,b}, Beck François^{c,d}, Spilka Stanislas^{e,f}

a Université Paris Descartes, Laboratoire de Psychopathologie et Processus de Santé, Institut de Psychologie, Sorbonne Paris Cité, France.

b Centre Pierre Nicole, « Consultation Jeunes Consommateurs », Croix-Rouge Française, 75005 Paris, France.

c Institut national de la statistique et des études économiques, Montrouge, France

d CESP, Faculté de médecine, Université Paris Sud, Faculté de médecine UVSQ, INSERM, Université Paris-Saclay, Villejuif, France

e Observatoire français des drogues et des toxicomanies, Saint-Denis, France

f CESP, Faculté de médecine, Université Paris Sud, Faculté de médecine UVSQ, INSERM, Université Paris-Saclay, Villejuif, France

Correspondence concerning this article should be addressed to:

Céline Bonnaire
Université Paris Descartes
Institut de Psychologie
Laboratoire de Psychopathologie et Processus de Santé EA 4057
71 Avenue Édouard Vaillant
92100 Boulogne-Billancourt
France
Tel: +33 1 76 53 29 52
celine.bonnaire@parisdescartes.fr

Abstract

Purpose: Patterns of psychostimulant (PST) use, psychological and physical factors, and family relationships were investigated in a nationally representative sample of adolescents. The differences between users of cocaine (Cc), ecstasy/MDMA, and amphetamine (EA) were analyzed, taking into account gender differences.

Methods: Substance use, psychological factors (lifetime suicide attempts, suicidal thoughts in the previous year, visits to a mental health professional), physical factors (health and body shape perceptions (BSP), body mass index (BMI)), and family relationships (existence of disagreements with parents, relationships with mother and father) were investigated in 26,351 17-year-old French adolescents.

Results: Lifetime suicidal behavior was strongly associated with EA use only. Suicidal ideation in the previous year was associated with Cc use amongst both boys and girls and associated with EA use amongst girls. Compared to Cc users, more EA users were overweight, while more Cc users were underweight compared to EA users. BSP, BMI, and EA use were related in girls only. Having a good or very good relationship with both mother and father was negatively associated with Cc use in boys and girls. Girls who had a good or very good relationship with their fathers seemed to be less prone to use EA. Boys with a good or very good relationship with their mothers seemed to be less prone to use EA.

Conclusion: The results of the present study could serve as a basis for prevention and harm reduction strategies/programs targeting youth use and for the adaptation of these strategies/programs according to substance and gender.

Key words: psychostimulants, cocaine, ecstasy/MDMA, amphetamine, adolescents

1. Introduction

As in other countries (Wu et al., 2010b; Jenkinson et al., 2014), the rates of use of psychostimulants (PST) among French adolescents have considerably varied over the past three decades. Though relatively low and stable between 1990 and 2010, the rates have increased significantly since 2010. Indeed, in 17-year-olds, cocaine (Cc) experimentation has increased from 0.9% in 2000 to 3.2% in 2014 (Beck et al., 2015). Since the early 1990s, the availability of stimulant substances, such as synthetic drugs (ecstasy, amphetamines), has increased in France (Beck, 2016). The proportion of current users of MDMA/ecstasy has increased from 0.3% in 2010 to 0.9% in 2014 and has now reached its highest level. Indeed, there is greater availability of MDMA/ecstasy, in both powder and tablet forms, with bigger and higher-dosed tablets than in the early 2000s (Spilka et al., 2015). MDMA/ecstasy use is ahead of Cc use among 18-25 year olds. MDMA/ecstasy experimentation has also increased in recent years among 17-year-olds, from 1.9 % in 2011 to 3.8% in 2014. Lifetime use of amphetamine is on the increase among 17-year-olds, from 2.4% in 2011 to 2.8% in 2014 (Spilka et al., 2015).

Due to the potential neurotoxic effects of adolescent MDMA abuse (Buchert et al., 2001), it is important to better understand the associated risks and protective factors. It is particularly important to understand those affecting use onset, as it is known to be associated with later misuse (Wu et al., 2010a). Based on the biopsychosocial model of addiction (Donovan and Marlatt, 1988; Skewes and Gonzalez, 2013), risk and protective factors could be linked to the substance and the substance effects (*i.e.*, biological factors), to the individual (*i.e.*, psychological factors), or to the user's environment (*i.e.*, social factors). Understanding the specific factors associated with initiation and use can improve efforts to target specific high-risk groups, delay or prevent initiation, and decrease regular use (Palamar and Kamboukos, 2014).

With regards to substance-related factors, the gateway theory, the common liability to substance use, or the route of administration model, have been used to understand how substance use starts and progresses in adolescents (Attaiaa et al., 2016; Fergusson et al., 2006; Kandel, 2002; Mayet et al., 2016). Several studies have found that initiation of ecstasy use can be predicted by an adolescent's early initiation of smoking, drinking, or cannabis use (Martins et al., 2006; Wu et al., 2010a; Zimmermann et al., 2005). Furthermore, adolescent users of drugs and alcohol are more likely to use cocaine in mid-adulthood (Fothergill et al., 2009), and most patients treated for methamphetamine misuse had used alcohol, marijuana, and

tobacco in early adolescence (Brecht et al., 2007). Further studies need to examine the role of tobacco, alcohol, and cannabis use in PST use onset, and the possible individual specificities for Cc, MDMA, and amphetamine. In France, it is forbidden to sell alcohol and tobacco to minors under the age of eighteen since 2009. French legislation also prohibits smoking in public places and therefore in those frequented by minors (sports or cultural centers) and in school grounds (public and private), including playgrounds.

With regards to psychological factors, several studies have shown associations between suicide — whether thoughts or attempts — and PST use (Kim et al., 2011; Wong et al., 2013; Doksat et al., 2017). Furthermore, adolescence is the developmental period with the highest risk for the emergence of eating disorders and associated weight-related behaviors (Martinsen and Sundgot-Borgen, 2013; Rohde et al., 2015). Substance use (tobacco, alcohol, and cocaine) has been linked to disordered eating and poor weight perception in adolescents (Antin and Paschall, 2011; Pasch et al., 2011; Eichen et al., 2012). However, to date, little research has examined either the association between perception of one's weight, body mass index (BMI), and PST use, or the possible specific impact of cocaine, MDMA, and amphetamine use.

Although adolescents generally spend most of their time with peers, parent-adolescent relationships are essential and serve as an adaptive and protective factor against youth substance use (Field et al., 2002; Wood et al., 2004; Piko and Kovács, 2010). Indeed, the quality of family relationships is a strong protective factor against the emergence of substance use disorders, especially in adolescents (Waldron et al., 2013; Steinberg, 2014; Strunin et al., 2015). Nevertheless, little is known about PST and most studies do not explore potential gender differences.

Thus, in light of all these findings, this study aimed to describe PST use, psychological factors (lifetime suicide attempts, suicidal thoughts in the previous year, visits to a mental health professional), physical factors (health and body self-perception, BMI), and family relationships (existence of disagreements with parents, relationships with mother and father) in relation to PST use in both boys and girls in a nationally representative sample of French adolescents. We also investigated gender differences regarding cocaine use on the one hand and ecstasy/MDMA and amphetamine use on the other hand.

2. Methods

2.1. Participants and procedure

The ESCAPAD Survey (Survey on health and behavior) is regularly conducted by the French Monitoring Centre for Drugs and Drug Addiction and provides national representative estimates of prevalence of drug use in adolescents. In France, all French adolescents must attend a one-day compulsory session of civic and military information after attaining the age of 17. It is at this occasion that adolescents (French nationals only) are invited to participate in the survey. ESCAPAD data collection takes place in March in 300 civilian or military centres across the French territory, where the one-day sessions take place. Regularly, there are on average 12% of absences or requests for postponement on any given day. Thus, the total number of teenagers summoned was estimated between 33,700 and 33,800 the year the data collection for this study took place. The anonymous pen and paper self-administered questionnaires follow the recommendations of the European Monitoring Centre for Drug and Drug Addiction (Bless et al., 1997). Though the population from the French overseas territories is also surveyed, the present study focuses on the mainland population (Beck et al., 2006). The survey obtained the Public Statistics general interest quality seal of approval from the National Council for Statistical Information. Participants are guaranteed full confidentiality and anonymity and can refuse to complete the form, as explicitly stated in the questionnaire guidelines. These conditions improve honest completion, with response rates superior to 99% (non-blank questionnaires/young people present) with respect to collection of sociodemographic data and substance use information.

The sample included 26,351 French adolescents aged 17. Responses were weighted to guarantee representativeness based on gender and age at the department level (i.e. administrative divisions of territory in France).

2.2. Measures

2.2.1. Sociodemographic variables

Gender, age, educational level (terms used: *pupil* for individuals in college or high school; *student* for individuals in higher education; and *trainee* for individuals in apprenticeship), grade repetition, parents who have separated or who live together, age when parents separated, and parents' socio-professional status were investigated.

2.2.2. Substance use

Tobacco, Alcohol, Cannabis: Adolescents were asked if they had already smoked tobacco, drunk alcohol, and smoked cannabis (and how old they were if they had), and whether they had done so in the past 12 months and the past 30 days.

Daily tobacco use was defined as smoking at least one cigarette per day and intensive tobacco use as more than 10 cigarettes per day. Daily alcohol use was defined as at least one alcoholic drink per day and regular alcohol use as more than 10 drinks in the past 30 days. Daily cannabis use was defined as at least one use per day. Regular cannabis use was defined as smoking cannabis more than 10 times per month.

The Cannabis Abuse Screening Test (CAST) was used to detect problematic use of cannabis. It is a 6-item scale screening for problematic patterns of cannabis use (Legleye et al., 2011; 2013) in the previous 12 months. Depending on the total score (from 0 to 24), users are classified as without risk (scores below 3), with a low risk (scores between 3 and 7), and at high risk for dependence (scores equal to or greater than 7).

Other illegal substance use: Adolescents were asked if they had ever used any of the following substances (at least once): hallucinogenic mushrooms, poppers, inhalation products, ecstasy/MDMA, amphetamine, LSD, crack, cocaine, heroin, ketamine, subutex/methadone, GHB, or any other drug.

Amphetamines refer to a group of molecules with related pharmacological effects (amphetamine, methamphetamine, and MDMA). Furthermore, because several national and international studies have found that many users typically consume both ecstasy and amphetamines (Pennay and More, 2010; Licht et al., 2012; Jenkinson et al., 2014), we grouped these two substances together in the analyses.

2.2.3. Psychological health

Lifetime suicide attempts: Adolescents were asked if they had ever tried to commit suicide and how old they were when they first tried.

Suicidal ideation in the previous year: Adolescents were asked if they had had suicidal thoughts in the previous year.

Visits to a mental health professional: Adolescents were asked if they had ever consulted a psychologist or a psychiatrist.

2.2.4. Physical health

Health self-perception was evaluated using the following question: “Compared to people your age, would you say that your health is “Not at all satisfactory/Very unsatisfactory, Unsatisfactory, Somewhat satisfactory, or Very satisfactory””.

Body shape perception (BSP) was evaluated using the following question: “How would you describe yourself physically? Much too thin, a little too thin, about the right weight, a little too fat, or much too fat.” By combining the higher and lower response categories, three categories were created for BSP: too thin, about the right weight, and too fat (Legleye et al., 2014). A similar question is often asked in studies about obesity and mental health to identify adolescents who "feel fat" (Jansen et al., 2008) or adolescents who perceive themselves as overweight (Perrin et al., 2010).

Body Mass Index (BMI) was calculated for all adolescents using their self-reported weight and height. Four categories were defined – a BMI between 18.5 and 24.9 kg/m² was considered normal, a BMI between 25 and 30 kg/m² was considered overweight, a BMI of equals to or greater than 30 kg/m² was considered obese, and a BMI below 18.5 kg/m² was considered underweight (McAdams et al., 2007; Dauphinot et al., 2009).

2.2.5. Family relationship

Three questions were asked to evaluate the quality of family relationships: “Has there ever been a time when you got along very badly with one of your parents?”; “How do you get along with your mother? Very well, well, reasonably, badly, or very badly”; and “How do you get along with your father? Very well, well, reasonably, badly, or very badly”. All answers were then categorized as “Good” (corresponding to the answers “very well” or “well”), “Bad” (corresponding to the answers “badly” or “very badly”), and “Reasonable” (corresponding to the answer “reasonably”).

2.3. Data analysis

Firstly, univariate analyses were conducted: Cocaine users (Cc) and Non-illicit drug users (NIU), Ecstasy/Amphetamine users (EA) and NIU, and PST users and NIU were compared. Next, gender differences were analyzed. A one-way analysis of variance was used to assess mean differences in continuous variables. For categorical data, differences in percentages were compared using the chi-squared test. A p-value < 0.05 was used as a test of significance

with a Bonferroni adjustment for multiple comparisons. To test the effect sizes, we calculated Cohen's d for the continuous variables and Phi-squared or Kramer's V for the categorical data.

Secondly, univariate analyses (logistic regressions) were conducted. Odds ratios (OR) and 95% confidence intervals (CI) were generated using logistic regressions to test for the unique contributions of socio-demographic data, substance use, physical health, mental health, and family relationships. Thirdly, adjusted odds ratios were calculated using multivariate logistic regressions to determine whether the univariate associations between PST use and all the variables investigated (physical and mental health, family relationships, and substance use) were maintained when adjusting for sociodemographic variables. Specifically, age, education, marital status, grade repetition, and having divorced parents were associated with PST status in univariate analyses. Finally, to test for the potentially moderating effects of gender, regressions were stratified by gender. These analyses were replicated for Cc on the one hand and EA on the other hand, in order to explore the specific characteristics of each of those substances. All analyses were performed using SPSS v. 20.

3. Results

3.1. PST users

NIU and PST users' characteristics are displayed in Table 1.

3.1.1. Substance use

Compared to NIU, more PST users had already smoked tobacco ($\phi_c = 0.09$, $p < 0.001$), and smoked daily ($\phi_c = 0.10$, $p < 0.001$) and intensively ($\phi_c = 0.07$, $p < 0.001$), had already drunk alcohol ($\phi_c = 0.03$, $p < 0.001$), and did so regularly ($\phi_c = 0.09$, $p < 0.001$) and daily ($\phi_c = 0.023$, $p = 0.005$), had already smoked cannabis ($\phi_c = 0.17$, $p < 0.001$), and did so regularly ($\phi_c = 0.11$, $p < 0.001$) and daily ($\phi_c = 0.08$, $p < 0.001$), and had a problematic use of cannabis and a higher total CAST score ($\phi_c = 0.11$, $p < 0.001$; $d = 0.58$, $p < 0.001$). All these effect sizes were small. PST users had used tobacco, alcohol, or cannabis at an earlier age and scored higher at the CAST ($d = 0.43$, $p < 0.001$; $d = 0.27$, $p < 0.001$; $d = 0.42$, $p < 0.001$; $d = 0.58$, $p < 0.001$ respectively). For boys, the strongest effect sizes were for tobacco at an earlier age ($d = 0.40$, $p < 0.001$) and CAST total score ($d = 0.50$, $p < 0.001$). For girls, the strongest effect sizes were for tobacco and cannabis at an earlier age ($d = 0.50$, $p < 0.001$, and $d = 0.45$, $p < 0.001$) and CAST total score ($d = 0.77$, $p < 0.001$)

In the total sample, in both boys and girls, PST use was associated with increased odds of tobacco use (lifetime (AOR = 12.66, $p < 0.001$ for boys; AOR = 20.99, $p < 0.001$ for girls) daily use (AOR = 4.30, $p < 0.001$ for boys; AOR = 4.55, $p < 0.001$ for girls), and intensive use (AOR = 4.01, $p < 0.001$ for boys; AOR = 3.48, $p < 0.001$ for girls)), alcohol use (lifetime (AOR = 11.66, $p = 0.015$ for boys; AOR = 3.69, $p = 0.012$ for girls), regular (AOR = 4.12, $p < 0.001$ for boys; AOR = 5.49, $p < 0.001$ for girls)), and cannabis use (lifetime (AOR = 13.12, $p < 0.001$ for boys; AOR = 29.48, $p < 0.001$ for girls), regular (AOR = 5.65, $p < 0.001$ for boys; AOR = 13.24, $p < 0.001$ for girls), and daily (AOR = 5.07, $p < 0.001$ for boys; AOR = 11.38, $p < 0.001$ for girls)). For girls, daily use of alcohol and cannabis problematic use were strongly associated with PST use (AOR = 11.16, $p = 0.029$ and AOR = 5.33, $p < 0.001$ respectively). In the total sample, in both boys and girls, early use of tobacco (AOR = 0.87, $p = 0.002$ for boys; AOR = 0.77, $p < 0.001$ for girls), alcohol (AOR = 0.90, $p = 0.001$ for boys; AOR = 0.86, $p < 0.001$ for girls), and cannabis (AOR = 0.82, $p = 0.007$ for boys; AOR = 0.66, $p < 0.001$ for girls) was negatively associated with PST use.

3.1.2. Physical and mental health

Compared to NIU, more PST girl users perceived their health as unsatisfactory or very unsatisfactory ($\phi_c = 0.02$, $p = .05$) with a small effect size. For both boys and girls, more PST users than NIU had already tried to commit suicide ($\phi_c = 0.07$, $p < 0.001$), had had suicidal ideation in the previous year ($\phi_c = 0.05$, $p < 0.001$), and had visited a mental health professional ($\phi_c = 0.03$, $p < 0.001$). All these effect sizes were small.

In the total sample, in both boys and girls, previous suicide attempts (AOR = 5.75, $p = 0.001$ for boys; AOR = 4.41, $p < 0.001$ for girls), suicidal ideation in the previous year (AOR = 1.87, $p = 0.044$ for boys; AOR = 3.55, $p < 0.001$ for girls), and visits to a mental health professional (AOR = 2.35, $p = 0.006$ for boys; AOR = 1.75, $p = 0.034$ for girls) were positively and strongly associated with PST use.

3.1.3. Family relationships

Compared to NIU, more PST users (total, boys and girls) had divorced parents ($\phi_c = 0.03$, $p = 0.001$), had undergone a period during which they had had a bad relationship with their parents ($\phi_c = 0.07$, $p < 0.001$) and had had a bad or very bad relationship with their fathers ($\phi_c = 0.05$, $p < 0.001$) and with their mothers ($\phi_c = 0.06$, $p < 0.001$). All these effect sizes were small.

In the total sample, in both boys and girls, going through a period of bad parent-adolescent relationship was positively associated with PST use (AOR = 3.35, $p < 0.001$ for boys; AOR = 2.66, $p < 0.001$ for girls). In girls, a good or very good relationship with the father inversely related to PST use (AOR = 0.36, $p < 0.001$), while in both boys and girls, a good or very good relationship with the mother was inversely related to PST use (AOR = 0.38, $p < 0.001$ for boys; AOR = 0.40, $p = 0.001$ for girls).

3.2. *Cc users*

Cc users' characteristics are displayed in Table 2. Unadjusted and adjusted odds ratio, confidence intervals and p -values and independent variables used in these analyses are displayed in Table 3.

3.2.1. *Substance use*

Compared to NIU, more Cc users smoked tobacco (daily ($\phi_c = 0.07$, $p < 0.001$) and intensively ($\phi_c = 0.04$, $p < 0.001$)), drank alcohol (lifetime ($\phi_c = 0.02$, $p = 0.033$) and regular use ($\phi_c = 0.07$, $p < 0.001$)), had already smoked cannabis ($\phi_c = 0.13$, $p < 0.001$), smoked regularly ($\phi_c = 0.10$, $p < 0.001$) and daily ($\phi_c = 0.07$, $p < 0.001$), and had a problematic use of cannabis ($\phi_c = 0.08$, $p < 0.001$). All these effect sizes were small. PST users had either used tobacco ($d = 0.33$, $p < 0.001$), alcohol ($d = 0.30$, $p = 0.002$), or cannabis consumption ($d = 0.33$, $p = 0.003$) at an earlier age and had higher CAST score ($d = 0.52$, $p < 0.001$).

For boys, Cc use was positively associated with increased odds of tobacco, alcohol, and cannabis use and negatively associated with age of first drink (AOR = 0.88, $p = 0.007$). The strongest associations were between lifetime use of tobacco and lifetime use of cannabis (AOR = 19.66, $p < 0.001$ and AOR = 31.00, $p < 0.001$ respectively) and between lifetime use of tobacco and daily and regular use of cannabis (AOR = 19.66, $p < 0.001$ and AOR = 31.00, $p < 0.001$ respectively).

For girls, Cc use was positively associated with increased odds of tobacco, alcohol, cannabis use and negatively associated with early use of tobacco and alcohol (AOR = 0.81, $p = 0.003$ and AOR = 0.86, $p = 0.009$ respectively). The strongest associations were between lifetime use of tobacco and cannabis (AOR = 29.76, $p = 0.001$ and AOR = 78.46, $p < 0.001$ respectively), daily use of alcohol (AOR = 25.55, $p = 0.004$). Problematic use of cannabis was associated with Cc use only in girls (AOR = 4.17, $p = 0.001$).

3.2.2. *Mental and physical health*

Compared to NIU, more Cc users were underweight and fewer Cc users had a normal weight ($\phi_c = 0.03$, $p = 0.013$). More Cc users had had suicidal ideation in the previous year ($\phi_c = 0.03$, $p < 0.001$). All these effect sizes were small. Among Cc users, more girls perceived their bodies as too fat and had had suicidal ideation in the previous year compared to boys.

For boys, Cc use was positively associated with increased odds of suicidal ideation in the previous year (AOR = 2.21, $p = 0.05$) and visits to a mental health professional (AOR = 2.36, $p = 0.05$). In girls, Cc use was positively associated with increased odds of suicidal ideation in the previous year (AOR = 2.95, $p = 0.001$)

3.2.3. Family relationships

Compared to NIU, more Cc users had divorced parents ($\phi_c = 0.02$, $p = 0.002$), had undergone a period during which they had had a bad relationship with their parents ($\phi_c = 0.04$, $p = 0.001$), and had had a bad or very bad relationship with their fathers ($\phi_c = 0.03$, $p < 0.001$) and with their mother ($\phi_c = 0.04$, $p < 0.001$). All these effect sizes were small.

For boys, Cc use was positively and strongly associated with a period of bad parent-adolescent relationship (AOR = 3.40, $p < 0.001$). In boys and girls, a good or very good parent-adolescent relationship (fathers and mothers) was negatively associated with Cc use (AOR = 0.46, $p < 0.001$ and AOR = 0.30, $p = 0.033$; AOR = 0.49, $p = 0.032$ and AOR = 0.38, $p = 0.012$ respectively).

3.3. A users

EA users' characteristics are displayed in Table 2. Unadjusted and adjusted odds ratio, confidence intervals and p -values and independent variables used in these analyses are displayed in Table 4.

3.3.1. Substance use

Compared to NIU, more EA users had already smoked tobacco ($\phi_c = 0.06$, $p < 0.001$) and smoked daily and intensively ($\phi_c = 0.06$, $p < 0.001$ and $\phi_c = 0.05$, $p < 0.001$ respectively), drank regularly ($\phi_c = 0.06$, $p < 0.001$), had already smoked cannabis ($\phi_c = 0.09$, $p < 0.001$), smoked regularly ($\phi_c = 0.04$, $p < 0.001$) and had a problematic use of cannabis ($\phi_c = 0.07$, $p = 0.026$). All these effect sizes were small.

EA users had either used tobacco ($d = 0.52$, $p < 0.001$), alcohol ($d = 0.24$, $p = 0.027$), or cannabis at an earlier age ($d = 0.42$, $p = 0.001$) and had higher CAST score ($d = 0.55$, $p < 0.001$).

In boys, EA use was associated with increased odds of tobacco use (especially lifetime (AOR = 8.47, $p < 0.001$), daily (AOR = 4.32, $p < 0.001$)) and age at first use (AOR = 0.77, $p = 0.001$), alcohol use (especially regular use (AOR = 3.55, $p < 0.001$) and age at first use (AOR = 0.90, $p = 0.039$)), and cannabis use (especially lifetime (AOR = 6.83, $p < 0.001$), regular (AOR = 5.22, $p < 0.001$) and age at first use (AOR = 0.78, $p = 0.014$)).

In girls, EA use was associated with increased odds of tobacco use (especially lifetime (AOR = 12.86, $p < 0.001$), intensive (AOR = 5.47, $p < 0.001$)) and age at first use (AOR = 0.81, $p = 0.001$), alcohol use (especially regular (AOR = 6.02, $p < 0.001$) and age at first use (AOR = 0.89, $p = 0.050$)), and cannabis use (especially lifetime (AOR = 12.89, $p < 0.001$), and age at first use (AOR = 0.66, $p = 0.007$)).

3.3.2. Mental and physical health

Compared to NIU, more EA users had already tried to commit suicide ($\phi_c = 0.08$, $p < 0.001$), had had suicidal ideation in the previous year ($\phi_c = 0.03$, $p < 0.001$), and had visited a mental health professional ($\phi_c = 0.02$, $p = 0.004$). All these effect sizes were small. Among EA users, more girls perceived their body as too fat, had already tried to commit suicide, and had had suicidal ideation in the previous year.

In boys, EA use was associated with increased odds of previous suicide attempt (AOR = 10.13, $p < 0.001$), and visits to a mental health professional (AOR = 2.58, $p = 0.032$).

In girls, EA use was associated with increased odds of body shape perception as too fat (AOR = 2.01, $p = 0.038$), overweight (AOR = 2.88, $p = 0.023$), previous suicide attempt (AOR = 10.80, $p < 0.001$), suicidal ideation in the previous year (AOR = 4.36, $p < 0.001$), and visits to a mental health professional (AOR = 2.21, $p = 0.048$).

3.3.3. Family relationships

Compared to NIU, more EA users had undergone a period during which they had had a bad relationship with their parents ($\phi_c = 0.06$, $p < 0.001$) and had had a bad or very bad relationship with their fathers ($\phi_c = 0.03$, $p = 0.001$) or their mothers ($\phi_c = 0.04$, $p < 0.001$). All these effect sizes were small. Among EA users, more girls than boys had undergone a

period during which they had had a bad relationship with their parents, and had had a bad or very bad relationship with their father.

In boys, EA use was associated with increased odds of a period of bad parent-adolescent relationship (AOR = 3.61, $p < 0.001$), and a good or very good relationship with the mother (AOR = 0.45, $p = 0.050$).

In girls, EA use was associated with increased odds of a period during which the relationship with parents was bad (AOR = 4.84, $p < 0.001$), and a good or very good relationship with the father (AOR = 0.23, $p < 0.001$).

3.4. EA users vs CC users

Compared to EA users, more Cc users smoked cannabis in their life ($\phi_c = 0.26$, $p < 0.001$), daily ($\phi_c = 0.14$, $p = 0.037$) and regularly ($\phi_c = 0.14$, $p = 0.043$). Compared to Cc users, more EA users were overweight ($\phi_c = 0.23$, $p = 0.010$) and had already tried to commit suicide ($\phi_c = 0.18$, $p = 0.008$). Conversely, more Cc users than EA users were underweight ($\phi_c = 0.23$, $p = 0.010$). Compared to EA users, parents of Cc users divorced at an earlier date ($d = 0.52$, $p = 0.032$).

4. Discussion

The present study evaluated the associations between psychostimulant use (cocaine, ecstasy, and amphetamine) and use of tobacco, alcohol, and cannabis, as well as associations with mental health, physical health, and family factors in a large population-based survey in France. We also investigated gender differences in these associations. Indeed, an understanding of the specific factors associated with initiation and use of psychostimulants can serve as a basis for targeting specific high-risk groups in order to delay initiation and decrease regular use (Palamar and Kamboukos, 2014). Several important findings were obtained: (1) PST users present specific patterns of substance use (tobacco, alcohol and cannabis), mental health problems, and family relationship and gender specificities; (2) Among PST users, Cc users and EA users present their own specific characteristics and factors associated with Cc use are different from those associated with EA use. They also differ depending on gender.

4.1. Substance use

Our findings regarding PST use confirm its strong association with tobacco, alcohol and cannabis use (Fergusson et al., 2006; Kandel, 2002; Macleod et al., 2004; Mayet et al., 2016; Wu et al., 2010a). Our results are in line with the common liability to substance use theory that is blind to the choice of the substance used first (Mayet et al., 2016). Indeed, unlike the order of initiations, the number of substances previously initiated, understood as a common liability of each of them, is important and increased the likelihood of subsequent use. This was previously emphasised by Attaiaa et al. (2016). Like in previous studies (Wu et al., 2010a; Martins et al., 2006; Zimmerman et al., 2005), we found that early use onset of tobacco, alcohol and cannabis is associated with initiation of PST use. Nevertheless, we found some gender and substance use particularities. Indeed, these associations are generally stronger in girls, especially in Cc users. Furthermore, Cc users are more frequently regular and daily users of cannabis (compared to EA users) and problematic use of cannabis is only associated with Cc use in girls,. In girls, Cc use is also associated with early onset use of tobacco and daily use of alcohol (the risk is increased by 25). In EA users, intensive use of tobacco and regular cannabis use are only associated with EA use in boys. On the one hand, these results suggest that prevention programs should take gender specificities into account as well as the type of substance used by individuals. On the other hand, it is necessary to further explore the associations between different substance uses. For example, daily alcohol use may be associated with Cc use in girls. Secondly, simultaneous use of multiple substances may lead to an additive 'high'. Thirdly, cocaine may be used for its antagonist effect on alcohol inebriation and to antagonize the deficit in psychomotor performance (Pennings et al., 2002). Further longitudinal investigation is needed to confirm these hypotheses.

4.2. Mental and physical health

One important finding with a medium effect size is that more EA users were overweight compared to Cc users while more Cc users were underweight compared to EA users. Furthermore, in girls, EA use was related to BSP (perceiving oneself as too fat) and BMI (being overweight). Thus, as highlighted by Eichen et al. (2012), the potential relationship between eating behaviors and weight and body perceptions and use of specific substances should be investigated, especially in EA female users. The different results that arose suggest that there may be different mechanisms at play that should be examined to establish more efficient prevention programs. As shown by previous studies on cannabis, alcohol, cigarettes, and anabolic steroids (Garry et al., 2003; Vidot et al., 2016), it would be interesting to further investigate the relationship between ecstasy or amphetamine use and weight loss tactics in this

age group. Furthermore, as highlighted by Zeller et al. (2016), pediatric weight management and bariatric surgery programs should screen for substance use, especially among girls.

Several studies have found a relationship between cocaine or ecstasy use and suicidal thoughts and behaviors in adolescents (Fernando et al., 2012; Garlow and Heninger, 2007; Kelly et al., 2002; Kim et al., 2011; Wong et al., 2013). However, in our study, lifetime suicidal behavior was strongly associated with EA use only (not Cc use) and suicidal ideation in the previous year was associated with Cc use in both boys and girls and EA use in girls. Though quite large, sample size may account for the small number of cases in the sample. Nevertheless, previous studies investigated different samples: clinical population samples who had completed suicide or made a suicide attempt (Fernando et al., 2012; Garlow and Heninger, 2007; Kelly et al., 2002) and general population samples (Kim et al., 2011; Wong et al., 2013). Thus, the association between substance use and suicidal behaviors may be different between samples from the general population and samples of individuals with a mental disorder, but also differ regarding substance use and gender. In previous studies, suicide ideation and attempts have been associated with perceived overweight status among adolescents (Eaton et al., 2005; Whetstone et al., 2007). The results of the present study could suggest that these associations are especially true for girls and that there is a need to further explore the association between body/weight perception, desire to lose weight, suicidal thoughts/attempts, and ecstasy or amphetamine use in girls.

4.3. Family relationships

Our findings are consistent with results from previous studies, indicating that certain parental characteristics (like living with both parents (Barrett and Turner, 2006)) and parental behaviors may deter adolescents from using substances (Piko and Kovács, 2010; Stone et al., 2012). However, unlike other previous studies on ecstasy use (Wu et al., 2010a; Palamar and Kamboukos, 2014), our study did not show any association between parental divorce and EA use. Parental divorce was only associated with Cc use. Moreover, the quality of the relationship with the parents seemed important. Having a good or very good relationship with mothers and fathers was negatively associated with Cc use in both boys and girls. Girls who had a good or very good relationship with their fathers seemed to be less prone to EA use, while it was with the mothers in boys. It is now necessary to better understand the positive or negative role of parent-adolescent relationships in adolescents' substance use, as well as the gender specificities.

This study has several limitations. First, the cross-sectional nature of the study did not enable us to determine the direction of the association between variables. Furthermore, despite the overall large sample size, there were very few girls in some categories. Thus, some statistical analyses were impossible ($n < 5$). As highlighted by Maning et al. (2001), there are concerns about the accuracy of some self-report studies, particularly those carried out in a formal educational context. This was not the case for this survey as it was not related to schooling, nevertheless there is a possibility that both over-reporting and under-reporting occurred. Finally, we made a methodological decision by grouping amphetamine and ecstasy use ($n=61$ and $n=40$ respectively). Nevertheless, amphetamines are sometimes used in different ways than MDMA (oral use for MDMA vs. possibility of injection for amphetamine/methamphetamine). Thus, the consequences of the uses of these two substances may differ and one might think that factors associated with their initiation are different. Further research needs to explore the potential specificities of these two kinds of substances.

Despite these limitations, the study sheds light on some important questions. Specific substance use associations (tobacco, alcohol, and cannabis), physical and mental health, and family factors were identified in the initiation and use of Cc and Ecstasy/Amphetamine. Thus, our results are in line with the biopsychosocial model of addiction which highlights the multicausality of illicit drug use. Indeed, factors implicated in illicit drug use are multiple and the initiation sequence cannot be reduced to the use of cannabis (preceded by alcohol and tobacco). Special attention needs to be given to this age group because of the deleterious impact of these substances on adolescent brains, especially that of MDMA (Parrott, 2001; Montoya et al., 2002). The results of this study can serve as a basis for prevention and harm reduction strategies/programs targeting youth use and to adapt these strategies/programs according to substance and to gender. Furthermore, given the social aspect of the initiation to illicit drugs in adolescence (i.e. affiliation with peers who use drugs) (Otten, et al., 2017), prevention programs need to attend to the peer system.

Declarations of interest: none

References

Attaiaa L.-A., Beck F., Richard J.-B., Marimoutou C., Mayet A., 2016. Relationships between substance initiation sequence and further substance use: A French cross-sectional nationwide study, *Addictive Behaviors* 57, 1-5.

Antin, T.M. Paschall, M.J., 2011. Weight perception, weight change intentions, and alcohol use among young adults. *Body Image*, 8(2), 149-156.

Barrett, A.E., Turner, R.J. 2006. Family structure and substance use in adolescence and early adulthood: examining explanations for the relationship. *Addiction* 101,109-120.

Beck, F. (dir.), 2016. *Young people and Addictions*, OFDT, Paris, 206 p.

Beck, F., Costes, J.-M., Legleye, S., Spilka S., 2006. The Escapad Survey on Drug Use: an original data collection tool on a sensitive topic. In: Lavallée, P., Rivest, L.-P. (Eds.), *Survey Methods: European and North American Practices* (pp. 56-59). Paris: Dunod.

Beck, F., Richard, J.B., Guignard, R., Le Nézet, O., Spilka S., 2015. Levels of drug use in France in 2014. *Tendances* 99, 8 p. <https://en.ofdt.fr/BDD/publications/docs/eftafbv3.pdf>

Bless, R., Korf, D., Riper, H., Diemel, S., 1997. Improving the Comparability of General Population Surveys on Drug Use in the European Union, Final Report. European Monitoring Centre for Drugs and Drug Addiction, Lisbon.

Brecht, M-L., Greenwell, L., Anglin M.D., 2007. Substance use pathways to methamphetamine use among treated users. *Addictive Behaviors* 32, 24-38.

Buchert, R. Obrocki, J., Thomasius, R., Vaterlein, O., Petersen, K., Jenicke, L., Bohuslavizki, K.H., Clausen, M., 2001. Long-term effects of ‘ecstasy’ abuse on the human brain studied by FDG PET. *Nuclear Medicine Communications* 22, 889-897.

Dauphinot, V., Wolff, H., Naudin, F., Guéguen, R., Sermet, C., Gaspoz, J.M., Kossovsky, M.P., 2009. New obesity body mass index threshold for self-reported data. *Journal of Epidemiology and Community Health* 63(2), 128-32.

Doksat, N.G., Zahmacioglu, O., Demirci, A.C., Kocaman, G.M., Erdogan, A. 2017. Association of suicide attempts and non-suicidal self-injury behaviors with substance use and family characteristics among children and adolescents seeking treatment for substance use disorder. *Substance Use and Misuse* 52(5), 604-613.

Donovan, D.M., Marlatt, G.A., 1988. The Guilford behavioral assessment series. Assessment of addictive behaviors. Guilford Press, New York, NY.

Eaton, D.K., Lowry, R., Brener, N.D., Galuska, D.A., Crosby, A.E. 2005. Associations of Body Mass Index and Perceived Weight With Suicide Ideation and Suicide Attempts Among US High School Students. *Archives of Pediatrics & Adolescent Medicine* 159(6), 513-519

Eichen, D.M., Conner, B.T., Daly, B.P., Fauber, R.L., 2012. Weight perception, substance use, and disordered eating behaviors: Comparing normal weight and overweight high-school students. *Journal of Youth and Adolescence* 41(1), 1-13.

Fergusson, D.M., Boden, J.M., Horwood, L., 2006. Cannabis use and other illicit drug use: testing the cannabis gateway hypothesis. *Addiction* 101, 556-569.

Fernando, T., Gilbert, J.D., Carroll, C.M., Byard, R.W. 2012. Ecstasy and suicide. *Journal of Forensic Sciences*, 57(4), 1137-1139.

Field T., Diego M., Sanders C., 2002. Adolescents' parent and peer relationships *Adolescence* 37, 121-130

Fothergill, K.E., Enseminger, M.E., Robertson, J.A., Juon, H.S., 2009. Pathways to adult marijuana and cocaine use: a prospective study of African Americans from age 6 to 42. *Journal of Health and Social Behavior* 50, 65-81.

Garlow, S.J., Heninger, M. 2007. Cocaine and alcohol use preceding suicide in African American and white adolescents. *Journal of Psychiatric Research* 41(6), 530-536.

Garry, J.M., Morrissey, S.L., Whetstone, L.M. (2003). Substance use and weight loss tactics among middle school youth. *International Journal of Eating Disorder* 33(1), 55-63.

Jenkinson, R., Jolley, D., Dietze, P., 2014. 'Weekend on the town': discrete sessions of drug use for a sample of young psychostimulant users. *Drug and Alcohol Review* 33, 428-435.

Jansen, W., van de Looij-Jansen, P.M., de Wilde, E.J., Brug, J. (2008) Feeling fat rather than being fat may be associated with psychological well-being in young Dutch adolescents. *Journal of Adolescent Health* 42, 128-136.

Kandel, D.B., 2002. Stages and pathway of drug involvement: examining the gateway hypothesis. Cambridge University Press, New York, NY.

Kelly, T.M., Cornelius, J.R., Lynch, K.G. 2002. Psychiatric and substance use disorders as risk factors for attempted suicide among adolescents: a case control study. *Suicide and Life-Threatening Behavior* 32(3), 301-3012.

Kim, J., Fan, B., Liu, X., 2011. Ecstasy use and suicidal behavior among adolescents: findings from a national survey. *Suicide and Life-Threatening Behavior* 41(4), 435-444.

Legleye, S., Beck, F., Spilka, S., Chau, N., 2014. Perception and Socioeconomic Status in Adolescent Self-Report Surveys. *Plos One* 9(5), e96768.

Legleye, S., Piontek, D., Kraus, L., 2011. Psychometric properties of the Cannabis Abuse Screening Test (CAST) in a French sample of adolescents. *Drug and Alcohol Dependence*, 113(2-3), 229-235.

Licht, C.L., Christoffersen, M., Okholm, M., Damgaard, L., Fink-Jensen, A., Knudsen, G.M., Erritzoe, D. 2012. Simultaneous polysubstance use among Danish 3,4-methylenedioxymethamphetamine and hallucinogen users: combination patterns and proposed biological bases. *Human Psychopharmacology: Clinical and Experimental* 27(4), 352-363.

Macleod, J., Oakes, R., Copello, A., Crome, I., Egger, M., Hickman, M., Oppekowski, T., Stokes-Lampard, H., Smith, G.D. 2004. Psychological and social sequelae of cannabis and other illicit drug use by young people: a systematic review of longitudinal, general population studies. *Lancet* 363, 1579-1588.

Manning, V., Best, D., Rawaf, S., Rowley, J., Floyd, K., Strang, J., 2001. Drug use in adolescence: the relationship between opportunity, initial use and continuation of use of four illicit drugs in a cohort of 14-16 years-olds in South London. *Drugs: Education, Prevention and Policy* 8(4), 397-405.

Martins, S.S., Mazzotti, G., Chilcoat, H.D., 2006. Recent-onset ecstasy use: association with deviant behaviors and psychiatric comorbidity. *Experimental and Clinical Psychopharmacology* 14, 275-286.

Martinsen, M., Sundgot-Borgen, J., 2013. Higher prevalence of eating disorders among adolescent elite athletes than controls. *Medicine and Science in Sports and Exercise* 45(6), 1188–1197.

- Mayet, A., Legleye, S., Beck, F., Falissard, B., Chau, N., 2016. The Gateway hypothesis, Common liability to addictions or the Route of administration model? An original modelling process linking the three theories. *European Addiction Research* 22(2), 107-117.
- McAdams, M.A., Van Dam, R.M., Hu, F.B., 2007. Comparison of Self-reported and Measured BMI as Correlates of Disease Markers in U.S. Adults. *Obesity*, 15(1), 188-196.
- Montoya, A.G., Sorrentino, R., Lukas, S.E., Price, B.H. 2002. Long-term neuropsychiatric consequences of "Ecstasy" (MDMA): a review. *Harvard Review of Psychiatry* 10, 212-220.
- Otten, R., Jung Mun, C, Dishion, T.J., 2017. The social exigencies of the gateway progression to the use of illicit drugs from adolescence into adulthood. *Addictive Behaviors* 73, 144-150.
- Palamar, J.,J., Kamboukos, D. 2014. An Examination of Sociodemographic Correlates of Ecstasy Use Among High School Seniors in the United States. *Substance Use and Misuse* 49, 1774-1783.
- Parrott, A.C. 2001. Human psychopharmacology of Ecstasy (MDMA): a review of 15 years of empirical research. *Human Psychopharmacology* 16, 557-577.
- Pasch, K.E., Klein, E.G., Laska, M.N., Velazquez, C.E., Moe, S.G., Lytle, L.A., 2011. Weight misperception and health risk behaviors among early adolescents. *American Journal of Health Behavior* 35(6), 797-806.
- Pennay, A., Moore, D. 2010. Exploring the micro-politics of normalisation: Narratives of pleasure, self-control and desire in a sample of young Australian 'party drug' users. *Addiction Research & Theory* 18(5), 557-571.
- Pennings, Ed.J.M., Leccese, A.P., de Wolff, F.A., 2002. Effects of concurrent use of alcohol and cocaine. *Addiction* 97(7), 773-783.
- Perrin, E.M., Boone-Heinonen, J., Field, A.E., Coyne-Beasley, T., Gordon-Larsen, P. (2010) Perception of overweight and self-esteem during adolescence. *International Journal of Eating Disorders* 43, 447-454.
- Piko, B.F., Kovács, E., 2010. Do parents and school matter? Protective factors for adolescents substance use. *Addictive Behaviors* 35, 53-56.

Rohde, P., Stice, E., Marti, C.N., 2015. Development and predictive effects of eating disorder risk factors during adolescence: Implications for prevention efforts. *International Journal of Eating Disorders* 48(2), 187-98.

Skewes, M.C., Gonzalez, V.M., 2013. The biopsychosocial model of addiction. In: Miller, P.M., Ball, S.A., Bates, M.E., Blume, A.W., Kampman, K.M., Kavanagh, D.J., Larimer, M.E., Petry, N.M., De Witte, P. (Eds), *Comprehensive addictive behaviors and disorders*, Vol. 1: Principles of addiction (pp. 61-70). Publisher: Elsevier Academic Press, San Diego, CA.

Spilka S., Le Nézet O., Ngantcha M., Beck F., 2015. Drug use in 17-year-olds: analysis of the ESCAPAD survey. *Tendances* 100, 8 p.
<https://en.ofdt.fr/BDD/publications/docs/eftassv5.pdf>

Steinberg, L., 2014. *Adolescence*, tenth ed. Mc Graw Hill Education, New York.

Stone, A.L., Becker, L.G., Huber, A.M., Catalano, R.F., 2012. Review of risk and protective factors of substance use and problem use in emerging adulthood. *Addictive Behaviors* 37(7), 747-775.

Strunin, L., Diaz-Martinez, L.R., Heeren, T., Winter, M., Kuranz, S., Hernández-Ávila C.A., Fernández-Varela, H., Solis-Torres, C., 2015. Parental monitoring and family relations: associations with drinking patterns among male and female Mexican students. *Addictive Behaviors* 51, 143-151.

Vidot, D.C., Prado, G., Hlaing, W.M., Florez, H.J., Arheart, K.L., Messiah, S.E., 2016. Metabolic Syndrome Among Marijuana Users in the United States: An Analysis of National Health and Nutrition Examination Survey Data. *The American Journal of Medicine* 129, 173-179.

Waldron, H.B., Brody, J.L., Robbins, M.S., Alexander, J.F., 2013. Functional family therapy for adolescent substance use disorders, in: Miller, P.M., Ball, S.A., Bates, M.E., Blume, A.W., Kampman, K.M., Kavanagh, D.J., Larimer, M.E., Nancy, PM., De Witte, P. (Eds), *Comprehensive addictive behaviors and disorders (Vol.3): interventions for addiction*. Elsevier Academic Press, San Diego, CA, US, pp. 109-116.

Whetstone, L.M., Morrissey, S.L., Cummings, D.M. 2007. Children at risk: the association between perceived weight status and suicidal thoughts and attempts in middle school youth. *Journal of School Health* 77(2), 59-66.

Wong, S.S., Zhou, B., Goebert, D., Hishinuma, E.S., 2013. The risk of adolescent suicide across patterns of drug use: a nationally representative study of high school students in the United States from 1999 to 2009. *Social Psychiatry and Psychiatric Epidemiology* 48, 1611-1620.

Wood M.D., Read J.P., Mitchell R.E., Brand N.H., 2004. Do parents still matter? Parent and peer influences on alcohol involvement among recent high school graduates. *Psychology of Addictive Behaviors* 18, 19-30

Wu, P., Liu, X., Fan B., 2010a. Factors associated with initiation of ecstasy use among US adolescents: findings from a national survey. *Drug and Alcohol Dependence* 106(2-3), 193-198.

Wu, P., Liu, X., Pham, T.H., Jin, J., Fan, B., Jin, Z., 2010b. Ecstasy use among US adolescents from 1999 to 2008. *Drug and Alcohol Dependence* 112, 33-38.

Zeller, M.H., Paugh, J., Wu, Y.P. 2016. Associations among excess weight status and tobacco, alcohol, and illicit drug use in a large national sample of early adolescent youth. *Prevention Science* 17(4), 483-492.

Zimmermann, P., Wittchen, H.U., Waszak, F., Nocon, A., Hofler, M., Lied, R., 2005. Pathways into ecstasy use: the role of prior cannabis use and ecstasy availability. *Drug and Alcohol Dependence* 79, 331-341.

Table 1: Sociodemographic, physical health, mental health, and family relationships of past year or lifetime non-illicit drug-users (except cannabis) and past year or lifetime PST users (psychostimulants only)

Variables	Non illicit drug users (NIU)							PST users only (PST)							p-Value ♂vs. ♀	Tot	NIU vs PST Boys	Girls
	Total n = 15205		Boys n = 8088		Girls n = 7104		p-Value ♂vs. ♀	Total n = 241		Boys n = 134		Girls n = 107						
	n/mean	%/SD	n/mean	%/SD	n/mean	%/SD			n/mean	%/SD	n/mean	%/SD	n/mean	%/SD				
<i>Sociodemographic characteristics</i>																		
Gender																		
Boys	8088	53.2						134	55.6						NS	NS	NS	
Girls	7104	46.8						107	44.4									
Education																		
Pupil	12440	86.8	6230	82.4	6204	91.7	<0.001	172	80.4	87	76.3	85	85.0	NA	0.013	NS	PST>NIU* PST>NIU*	
Student	472	3.3	269	3.6	203	3.0		8	3.7	3	2.6	5	5.0					
Trainee	1419	9.9	1059	14.0	359	5.3		34	15.9	24	21.1	10	10.0					
Repetition																		
Yes	5532	36.8	3361	42.2	2166	30.7	<0.001	129	54.4	77	59.2	52	48.6	0.06	<0.001	PST>NIU*** PST<NIU***	PST>NIU***	
No	9501	63.3	4607	57.8	4889	69.3		108	45.6	53	40.8	55	51.4					
Divorced parents																		
Yes	3526	23.2	2211	27.5	1978	27.9	NS	78	32.4	42	31.3	36	33.6	NS	0.001	PST>NIU**	PST>NIU***	
No	11679	76.8	5840	72.5	5100	72.1		163	67.6	92	68.7	71	66.4					
Age	8.8	5.0	9.0	5.1	8.6	5.0		8.5	4.7	9.0	4.7	7.9	4.6					NS
<i>Susbtance use characteristics</i>																		
Tobacco																		
Age at first cigarette	14.4	1.9	14.4	1.9	14.4	1.6	0.036	13.6	1.9	13.6	2.1	13.6	1.6	NS	<0.001	PST<NIU***	PST<NIU***	
Lifetime use																		
Yes	8803	58.1	4596	57.1	4201	59.3	0.007	231	95.9	127	94.8	104	97.2	NS	<0.001	PST>NIU***	PST>NIU*	
No	6342	41.9	3450	42.9	2886	40.7	0.007	10	4.1	7	5.2	3	2.8	NS	<0.001	PST<NIU***	PST<NIU***	
Daily use																		
Yes	3160	20.9	1812	22.5	1346	19.0	<0.001	133	55.4	74	55.6	59	55.1	NS	<0.001	PST>NIU***	PST>NIU***	
No	11977	79.1	6229	77.5	5738	81.0	<0.001	107	44.6	59	44.4	48	44.9	NS	<0.001			
Intensive use																		
Yes	646	4.3	433	5.4	213	3.0	<0.001	37	15.4	26	19.5	11	10.3	0.03	<0.001	PST>NIU***	PST>NIU***	
No	14491	95.7	7608	94.6	6871	97.0	<0.001	203	84.6	107	80.5	96	89.7	0.03	<0.001			
Alcohol																		
Age at first alcoholic drink	13.9	2.3	13.7	2.4	14.2	2.1	<0.001	13.3	2.1	13.1	2.3	13.6	1.8	NS	<0.001	PST<NIU**	PST<NIU**	
Lifetime user																		
Yes	13454	88.9	7233	89.9	6212	87.7	<0.001	234	97.1	131	97.8	103	96.3	NS	<0.001	NA	NA	
No	1685	11.1	811	10.1	871***	12.3		7	2.9	3	2.2	4	3.7					
Regular use																		
Yes	1149	7.7	889	11.3	260	3.7	<0.001	65	28.0	47	35.9	18	17.8	0.002	<0.001	PST>NIU***	PST>NIU***	
No	13695	92.3	6960	88.7	6723	96.3		167	72.0	84	64.1	83	82.2					
Daily use																		

Yes	96	0.6	87***	1.1	9	0.1	<0.001	5	2.2	4	3.1	1	1.0	NS	0.005	NA	NA
No	14748	99.4	7762	98.9	6974	99.9		227	97.8	127	96.9	100	99.0				
Cannabis																	
Age at first joint	15.6	1.2	15.5**	1.3	15.7	1.0	0.001	15.1	1.2	15.1	1.2	15.2	1.2	NS	<0.001	PST<NIU***	PST<NIU***
Lifetime use																	
Yes	4074	26.9	2377***	29.6	1696	23.9	<0.001	212	88.0	115	85.8	97	90.7	NS	<0.001	PST>NIU***	PST>NIU***
No	11058	73.1	5659	70.4	5388	76.1		29	12.0	19	14.2	10	9.3			PST<NIU***	
Regular use																	
Yes	377	2.5	318***	3.9	59	0.8	<0.001	42	17.5	29	21.6	13	12.3	0.04	<0.001	PST>NIU***	PST>NIU***
No	14792	97.5	7741	96.1	7038	99.2		198	82.5	105	78.4	93	87.7				
Daily use																	
Yes	161	1.1	140***	1.7	21	0.3	<0.001	19	7.9	13	9.7	6	5.7	NS	<0.001	PST>NIU***	PST>NIU***
No	15008	98.9	7919	98.3	7076	99.7		221	92.1	121	90.3	100	94.3				
CAST																	
Problematic use	261	9.8	213***	13.1	48	4.6	<0.001	115	47.7	65	48.5	50	46.7	NS	<0.001	PST>NIU***	PST>NIU***
Non-problematic use	2406	90.2	1416	86.9	990	95.4		126	52.3	68	51.5	57	53.3				
Total score	2.1	3.2	2.6***	3.6	1.3	2.4	<0.001	4.3	4.3	4.6	4.4	3.9	4.1	NS	<0.001	PST>NIU***	PST>NIU***
	median	range	median	range	median	range		median	range	median	range	median	range				
	1.00	0-24	1.00	0-24	0.00	0-19		3.50	0-22	4.00	0-22	2.50	0-18				
Physical Health characteristics																	
Health self-perception																	
Satisfactory /very satisfactory	14391	95.3	7694	95.9	6686	94.6		223	93.7	127	96.2	96	90.6				
Unsatisfactory/Very unsatisfactory	712	4.7	331	4.1	380***	5.4	<0.001	15	6.3	5	3.8	10	9.4	0.06	NS	NS	PST<NIU*
Body shape perception																	
Too thin	1722	11.5	1371***	17.2	348	5.0		30	12.7	24	18.3	6	5.7				
About the right weight	9456	63.1	5379	67.4	4068	58.2	<0.001	142	59.9	88	67.2	54	50.9	<0.001	NS	NS	NS
Too fat	3812	25.4	1235	15.5	2576***	36.8		65	27.4	19	14.5	46	43.4				
BMI																	
Obese	262*	1.8	147	1.9	115	1.7		1	0.4	1	0.8	0					
Overweight	1199	8.3	790***	10.2	409	6.1	<0.001	20	8.8	12	9.3	8	8.2	NA	NS	NS	NS
Normal	10865	75.3	5982	77.2	4876	73.1		179	78.9	104	80.6	75	76.5				
Underweight	2100	14.6	827	10.7	1271***	19.1		27	11.9	12	0.3	15	15.3				
Mental Health characteristics																	
Past suicide attempt																	
Yes	195	1.3	487	6.1	771***	11.0	<0.001	18	7.5	5	3.8	13	12.1	0.01	<0.001	PST>NIU***	PST>NIU***
No	14904	98.7	7507	93.9	6246	89.0		221	92.5	127	96.2	94	87.9				
Age at first attempt	14.3	2.7	13.7*	3.3	14.5	2.5	0.057	14.4	1.9	15.4	1.9	14.1	1.9	NS	NS	NS	NS
Suicidal ideation in the past year																	
Yes	1261	8.4	51	0.6	144***	2.0	<0.001	46	19.6	14	10.8	32	30.5	<0.001	<0.001	PST>NIU*	PST>NIU***
No	13762	91.6	7974	99.4	6917	98.0		189	80.4	116	89.2	73	69.5				
Mental health professional																	
Yes	1113	7.4	405	5.1	707***	10.0	<0.001	31	13.0	12	9.1	19	17.8	0.04	0.001	PST>NIU*	PST>NIU*
No	13953	92.6	7606	94.9	6335			208	87.0	120	90.9	88	82.2				
Family Relationships characteristics																	
Times of very bad relationship with parents																	
Yes	3180	21.7	1275	16.4	1902***	27.6	<0.001	106	45.7	50	39.1	56	53.8	0.02	<0.001	PST>NIU***	PST>NIU***

No	11488	78.3	6487	83.6	4992	72.4		126	54.3	78	60.9	48	46.2				PST<NIU***
Age	13.7	2.5	13.7	2.6	13.7	2.4	NS	14.0	2.2	14.2	2.3	13.8	2.0	NS	NS	NS	NS
Relationship with father																	
Good/very good	11830	83.2	6473	85.5	5347	80.6		151	68.9	89	74.8	62	62.0				PST<NIU***
Bad/very bad	676	4.8	285	3.8	391***	5.9	<0.001	20	9.1	10	8.4	10	10.0	NS	<0.001	PST>NIU**	PST>NIU***
Reasonable	1720	12.1	817	10.8	900***	13.6		48	21.9	20	16.8	28	28.0			PST>NIU**	PST>NIU***
Relationship with mother																	
Good/very good	13289	89.4	7095	90.1	6183	88.7		175	75.1	98	77.2	77	72.6				
Bad/very bad	374	2.5	167	2.1	207**	3.0	0.001	19	8.2	9	7.1	10	9.4	NS	<0.001	PST>NIU***	PST>NIU***
Reasonable	1197	8.1	611	7.8	584	8.4		39	16.7	20	15.7	19	17.9			PST>NIU***	PST>NIU***

Notes: n/mean = number of categorical variables and mean for continuous variables; %/SD = percentage for categorical variables and standard deviation for continuous variables; NA = not applicable (statistical conditions are not met n<5); * < 0.05, ** p< 0.01, *** p< 0.001; Given the heterogeneous scores at the CAST, we have added median and interquartile range

Table 2: Sociodemographic data, physical health, mental health, and family relationships of non-illicit drug users (except cannabis), cocaine users and ecstasy/amphetamine users

Variables	No illicit drug users (NIU)							Cocaine users (Cc)							Ecsatsy/Amphetamine users (EA)							p-Value CC vs EA	p-Value CC vs EA	p-Value CC vs EA	p-Value CC vs EA
	Total n = 15205		Boys n = 8088		Girls n = 7104		p- Valu e ♂vs. ♀	Total n = 120		Boys n = 68		Girls n = 52		p- Valu e ♂vs. ♀	Total n = 101		Boys n = 59		Girls n = 42						
	n/me an	%/S D	n/me an	%/S D	n/me an	%/S D			n/me an	%/S D	n/me an	%/S D	n/me an		%/S D		n/me an	%/S D	n/me an	%/S D	n/me an	%/S D	p- Valu e ♂vs. ♀	p- Valu e CC vs NIU	p- Valu e EA vs NIU
Sociodemographic characteristics																									
Gender																									
Boys	8088	53.2						68	56.7					NS	59	58.4					NS	NS		NS	
Girls	7104	46.8						52	43.3						42	41.6									
Education																									
Pupil	12440	86.8	6230	82.4	6204	91.7	<0.001	87	82.9	48	85.7	39	79.6	NA	72	78.3	37	69.8	35	89.7	NA	NA	0.05	NS	
Student	472	3.3	269	3.6	203	3.0		2	1.9	0	0	2	4.1		5	5.4	3	5.7	2	5.1					
Trainee	1419	9.9	1059	14.0	359	5.3		16	15.2	8	14.3	8	16.3		15	16.3	13	24.5	2	5.1					
Repetition																									
Yes	5532	36.8	3361	42.2	2166	30.7	<0.001	69	58.5	42	63.6	27	51.9	NS	47	47.5	30	52.6	17	40.5	NS	<0.001	0.018	CC>EA* CC<EA*	
No	9501	63.3	4607	57.8	4889	69.3		49	41.5	24	36.4	25	48.1		52	52.5	27	47.4	25	59.5					
Divorced parents																									
Yes	3526	23.2	5840	72.5	5100	72.1		42	35.0	23	33.8	19	36.5		28	27.7	17	28.8	11	26.2		0.002	NS	NS	
No	11679	76.8	2211	27.5	1978	27.9	NS	78	65.0	45	66.2	33	63.5	NS	73	72.3	42	71.2	31	73.8	NS				
Age	8.8	5.0	9.0	5.1	8.6.*	5.0	0.044	7.6	4.6	7.9	4.6	7.3	4.7	NS	10.0	4.7	10.7	4.7	8.9	4.7	NS	NS	NS	CC<EA*	
Substance use characteristics																									
Tobacco																									
Age at first cigarette	14.4	1.9	14.3	1.9	14.4	1.6	0.036	13.8	1.8	13.8	1.9	13.8	1.7	NS	13.4	2.0	13.3	2.4	13.5	1.5	NS	<0.001	<0.001	NS	
Lifetime use																									
Yes	8803	58.1	4596	57.1	4201	59.3		117	97.5	66	97.1	51	98.1	NA	94	93.1	54	91.5	40	95.2	NS	NA	<0.001	NS	
No	6342	41.9	3450	42.9	2886	40.7	0.007	3	2.5	2	2.9	1	1.9		7	6.9	5	8.5	2	4.8					

Daily use	Yes	3160	20.9	1812	22.5	1346	19.0	<0.0	65	54.6	36	53.7	29	55.8	NS	54	53.5	33	55.9	21	50.0	NS	<0.0	<0.0	NS
No	11977	79.1	6229	77.5	5738	81.0	01	54	45.4	31	46.3	23	44.2	47		46.5	26	44.1	21	50.0	01		01		
Intensive use	Yes	646	4.3	433	5.4	213	3.0	<0.0	16	13.4	10	14.9	6	11.5	NS	16	15.8	14	23.7	2	4.8	NA	<0.0	<0.0	NS
No	14491	95.7	7608	94.6	6871	97.0	01	103	86.6	57	85.1	46	88.5	85		84.2	45	76.3	40	95.2	01		01		
Alcohol	Age at first alcoholic drink	13.9	2.3	13.7	2.4	14.2	2.1	<0.0	13.2	2.3	12.9	2.4	13.7	2.1	0.059	13.4	1.9	13.3	2.1	13.6	1.7	NS	0.002	0.027	NS
Lifetime use	Yes	13454	88.9	7233	89.9	6212	87.7	<0.0	114	95.0	66	97.1	48	92.3	NA	100	99.0	58	98.3	42	100.0	NA	0.02	NA	NS
No	1685	11.1	811	10.1	871	12.3	01	6	5.0	2	2.9	4	7.7	1		1.0	1	1.7	0						
Regular use	Yes	1149	7.7	889	11.3	260	3.7	<0.0	33	28.4	25	36.8	8	16.7	0.014	27	28.1	19	33.9	8	20.0	NS	<0.0	<0.0	NS
No	13695	92.3	6960	88.7	6723	96.3	01	83	71.6	43	63.2	40	83.3	69		71.9	37	66.1	32	80.0	01		01		
Daily use	Yes	96	0.6	87	1.1	9	0.1	<0.0	4	3.4	3	4.4	1	2.1	NA	1	1.0	1	1.8	0	100.0	NA	NA	NA	NS
No	14748	99.4	7762	98.9	6974	99.9	01	112	96.6	65	95.6	47	97.9	95		99.0	55	98.2	40						
Cannabis	Age at first joint	15.6	1.2	15.4	1.2	15.7	1.0	0.001	15.2	1.2	15.1	1.3	15.4	1.1	NS	15.1	1.2	15.0	1.1	15.2	1.2	NS	0.003	0.001	NS
Lifetime use	Yes	4074	26.9	2377	29.6	1696	23.9	<0.0	114	95.0	64	94.1	50	96.2	NA	78	77.2	44	74.6	34	81.0	NS	<0.0	<0.0	CC>EA***
No	11058	73.1	5659	70.4	5388	76.1	01	6	5.0	4	5.9	2	3.8	23		22.8	15	25.4	8	19.0	01		01		
Regular use	Yes	377	2.5	318	3.9	59	0.8	<0.0	25	21.0	18	26.5	7	13.7	0.07	11	10.9	10	16.9	1	2.4	NA	<0.0	<0.0	CC>EA*
No	14792	97.5	7741	96.1	7038	99.2	01	94	79.0	50	73.5	44	86.3	90		89.1	49	83.1	41	97.6	01		01		
Daily use	Yes	161	1.1	140	1.7	21	0.3	<0.0	12	10.1	9	13.2	3	5.9	NA	3	3.0	3	5.1	0	100.0	NA	<0.0	NA	CC>EA*
No	15008	98.9	7919	98.3	7076	99.7	01	107	89.9	59	86.8	48	94.1	98		97.0	56	94.9	42				01		
CAST	Problematic use	261	9.8	213	13.1	48	4.6	<0.0	18	24.0	10	22.2	8	26.7	NS	11	19.0	9	27.3	2	8.0	NA	<0.0	0.026	NS
Non-problematic use	2406	90.2	1416	86.9	990	95.4	01	57	76.0	35	77.8	22	73.3	47		81.0	24	72.7	23	92.0	01				
Total score		2.1	3.2	2.6	3.6	1.3	2.4	<0.0	4.1	4.4	4.6	5.1	3.5	3.3	NS	3.9	3.3	4.9	3.7	2.6	2.2	0.008	<0.0	<0.0	NS

	median	range	median	range	median	range		median	range	median	range	median	range		median	range	median	range	median	range				
	1.00	0-24	1.00	0-24	0.00	0-19		3.00	0-22	4.00	0-22	2.50	0-11		3.00	0-18	5.00	0-18	2.00	0-8				
Physical Health characteristics																								
Health self-perception																								
Satisfactory /very satisfactory	14391	95.3	7694	95.9	6686	94.6	<0.001	5	4.2	67	100	46	90.2	NA	95	95.0	55	94.8	40	4.8	NA	NS	NS	NS
Unsatisfactory /Very unsatisfactory	712	4.7	331	4.1	380	5.4		113	95.8	0	0	5	9.8		5	5.0	3	5.2	2	95.2				
Body shape perception																								
Too thin	1722	11.5	1371	17.2	348	5.0	<0.001	16	13.6	14	20.9	2	3.9	<0.001	12	12.1	9	15.8	3	7.1	<0.001	NS	NS	NS
About the right weight	9456	63.1	5379	67.4	4068	58.2		72	61.0	44	65.7	28	54.9		59	59.6	41	71.9	18	42.9				
Too fat	3812	25.4	1235	15.5	2576	36.8		30	25.4	9	13.4	21	41.2		28	28.3	7	12.3	21	50.0				
BMI																								
Obese	262*	1.8	147	1.9	115	1.7		0		0		0			1	1.0	1	1.8	0					
Overweight	1199	8.3	790	10.2	409	6.1	<0.001	6	5.4	4	6.2	2	4.3	NA	12	12.5	6	10.5	6	15.4	NS	0.013	NA	CC<EA**
Normal	10865	75.3	5982	77.2	4876	73.1		7	6.3	58	89.2	40	87.0		67	69.8	42	73.7	25	64.1				
Underweight	2100	14.6	827	10.7	1271	19.1		98	88.3	3	4.6	4	8.7		16	16.7	8	14.0	8	20.5				CC>EA**
Mental Health characteristics																								
Past suicide attempt																								
Yes	195	1.3	487	6.1	771	11.0	<0.001	4	3.4	1	1.5	3	5.8	NA	13	13.0	4	6.9	9	21.4	0.034	NA	<0.001	CC<EA**
No	14904	98.7	7507	93.9	6246	89.0		115	96.6	66	98.5	49	94.2		87	87.0	54	93.1	33	78.6				
Age at first attempt	14.3	2.7	13.7	3.3	14.5	2.5	0.057	14.5	3.3	17.0	-	13.7	3.5	NA	14.5	1.6	15.0	2.0	14.3	1.4	NS	NS	NS	NS
Suicidal ideation in the past year																								
Yes	1261	8.4	51	0.6	144	2.0	<0.001	22	19.0	8	12.1	14	28.0	0.028	18	18.2	4	7.0	14	33.3	0.001	<0.001	0.001	NS
No	13762	91.6	7974	99.4	6917	98.0		94	81.0	58	87.9	36	72.0		81	81.8	53	93.0	28	66.7				

Mental health professional																								
Yes	1113	7.4	405	5.1	707	10.0		11	9.2	6	9.0	5	9.6		15	15.0	6	10.3	9	21.4				
No	13953	92.6	7606	94.9	6335		<0.001	108	90.8	61	91.0	47	90.4	NS	85	85.0	52	89.7	33	78.6	NS	NS	0.007	NS
Family Relationships characteristics																								
Times of very bad relationship with parents																								
Yes	3180	21.7	1275	16.4	1902	27.6		47	40.5	24	36.9	23	45.1		50	51.0	24	42.1	26	63.4				
No	11488	78.3	6487	83.6	4992	72.4	<.001	69	59.5	41	63.1	28	54.9	NS	48	49.0	33	57.9	15	36.6	0.030	<0.001	<.001	NS
Age	13.7	2.5	13.7	2.6	13.7	2.4	NS	14.1	2.1	14.1	2.2	14.1	2.1	NS	14.0	2.1	14.2	2.6	13.8	1.7	NS	NS	NS	NS
Relationship with father																								
Good/very good	11830	83.2	6473	85.5	5347	80.6	<0.001	77	70.0	42	70.0	35	70.0	NA	62	68.1	41	77.4	21	55.3	NS (0.07)	0.001	0.001	NS
Bad/very bad	676	4.8	285	3.8	391	5.9		8	7.3	6	10.0	2	4.0		8	8.8	4	7.5	4	10.5				
Reasonable	1720	12.1	817	10.8	900	13.6		25	22.7	12	20.0	13	26.0		21	23.1	8	15.1	13	34.2				
Relationship with mother																								
Good/very good	13289	89.4	7095	90.1	6183	88.7		87	75.0	51	78.5	36	70.6	NS	73	75.3	41	74.5	32	76.2				
Bad/very bad	374	2.5	167	2.1	207	3.0	0.001	8	6.9	2	3.1	6	11.8		9	9.3	6	10.9	3	7.1	NS	<0.001	<0.001	NS
Reasonable	1197	8.1	611	7.8	584	8.4		21	18.1	12	18.5	9	17.6		15	15.5	8	14.5	7	16.7				

Notes: n/mean = number of categorical variables and mean for continuous variables; %/SD = percentage for categorical variables and standard deviation for continuous variables; NA = not applicable (statistical conditions are not met n<5); * < 0.05, ** p< 0.01, *** p< 0.001; Given the heterogeneous scores at the CAST, we have added median and interquartile range

Table 3: Associations between physical and mental health, family relationships and substance use and Cocaine use in whole sample, boys and girls

	Total n = 120				Cocaine users							
	OR	95%CI	AOR	95%CI	OR	95%CI	AOR	95%CI	OR	95%CI	AOR	95%CI
Age	1.12	0.94,1.34			1.08	0.84,1.38			1.18	0.91,1.52		
Education												
Pupil	Reference				Reference				Reference			
Student	0.61	0.15,2.47			NA				1.57	0.38,6.53		
Trainee	1.61	0.94,2.76			0.98	0.46,2.08			3.54	1.64,7.64		
Doubling school	2.42	1.67,3.49			2.40	1.45,3.97			2.44	1.41,4.21		
Parents divorced	1.78	1.22,2.60			1.72	1.04,2.84			1.87	1.06,3.31		
Tobacco Life	28.10	8.93,88.42	22.95	7.26,72.48	24.77	6.06,101.20	19.66	4.78,80.92	35.04	4.84,253.67	29.76	4.09,216.09
Age of first cigarette	0.85	0.78,0.93	0.88	0.79,0.96	0.88	0.79,0.98	0.92	0.81,1.05	0.81	0.70,0.93	0.81	0.70,0.93
Tobacco Daily	4.56	3.17,6.56	4.30	2.86,6.44	3.99	2.46,6.47	4.45	2.53,7.82	5.37	3.10,9.32	4.22	2.36,7.55
Tobacco Intensive	3.48	2.05,5.93	3.00	1.61,5.58	3.08	1.56,6.08	2.97	1.27,6.93	4.21	1.78,9.96	3.49	1.42,8.59
Alcohol Life	2.38	1.04,5.42	3.07	1.13,8.39	3.70	0.90,15.13	NA		1.68	0.60,4.68	1.70	0.60,4.77
Age of first drink	0.90	0.84,0.96	0.88	0.82,0.94	0.89	0.82,0.97	0.88	0.80,0.97	0.91	0.81,1.03	0.86	0.77,0.96
Alcohol Regular	4.74	3.15,7.12	4.20	2.67,6.61	4.55	2.77,7.49	4.39	2.48,7.77	5.17	2.40,11.16	5.20	2.38,11.35
Alcohol Daily	5.49	1.98,15.17	2.83	0.68,11.85	4.12	1.27,13.35	1.68	0.23,12.40	16.49	2.05,132.74	25.55	2.85,229.32
Cannabis Life	51.57	22.67,117.29	45.03	19.69,102.97	38.09	13.86,104.72	31.00	11.16,86.14	79.42	19.30,326.74	78.46	18.97,324.50
Age of first joint	0.81	0.71,0.93	0.85	0.72,0.99	0.81	0.70,0.95	0.85	0.70,1.04	0.80	0.61,1.04	0.78	0.59,1.02
Cannabis Regular	10.43	6.64,16.41	6.79	3.90,11.83	8.76	5.05,15.19	6.25	3.16,12.36	18.98	8.21,43.86	12.38	4.60,33.28
Cannabis Daily	10.45	5.64,19.37	6.41	2.87,14.30	8.63	4.20,17.74	7.58	3.13,18.37	21.06	6.08,72.96	5.18	0.65,41.36
CAST Problematic use	2.91	1.67,5.02	1.55	0.78,3.08	1.90	0.93,3.89	0.89	0.34,2.37	7.50	3.17,17.71	4.17	1.53,11.36
Health self,perception												
Good /very satisfying	Reference		Reference		Reference		Reference		Reference		Reference	
Less/no satisfying	0.89	0.36,2.20	0.57	0.18,1.81	NA		NA		1.91	0.76,4.84	0.98	0.30,3.19
Body shape perception												
Too thin	1.22	0.71,2.10	1.03	0.55,1.92	1.25	0.68,2.28	1.09	0.54,2.22	0.83	0.20,3.52	0.78	0.18,3.31
About the right weight	Reference		Reference		Reference		Reference		Reference		Reference	
Too fat	1.03	0.67,1.58	1.03	0.66,1.63	0.89	0.43,1.83	0.81	0.36,1.83	1.18	0.67,2.09	1.06	0.59,1.92
BMI												
Obese	NA		NA	0.04,2.00	NA		NA		NA		NA	
Overweight	0.55	0.24,1.27	0.40	0.15,1.09	0.52	0.19,1.44	0.32	0.07,1.31	0.60	0.14,2.47	0.57	0.14,2.38
Normal	Reference		Reference		Reference		Reference		Reference		Reference	
Underweight	0.37	0.17,0.80	0.44	0.20,0.96	0.37	0.12,1.20	0.45	0.14,1.45	0.38	0.14,1.07	0.44	0.15,1.23
Committed suicide lifetime	2.66	0.97,7.28	1.97	0.61,6.33	2.37	0.32,17.40	2.66	0.36,19.88	2.94	0.91,9.54	1.49	0.35,6.32
Age of first suicide	1.02	0.71,1.48	0.90	0.60,1.34	1.37	0.77,2.45	NA		0.88	0.58,1.33	0.78	0.52,1.17
Suicidal ideation past year	2.55	1.60,4.08	2.66	1.63,4.33	2.13	1.01,4.48	2.21	0.99,4.94	3.15	1.69,5.87	2.95	1.56,5.59

See professional mental health	1.28	0.68,2.38	1.42	0.75,2.66	1.85	0.79,4.30	2.36	0.99,5.60	0.95	0.38,2.40	0.87	0.34,2.22
Period of very bad relationship with parents	2.46	1.69,3.57	2.34	1.55,3.54	2.98	1.79,4.95	3.40	1.91,6.02	2.16	1.24,3.75	1.57	0.86,2.87
Age of very bad relationship	1.07	0.94,1.22	1.08	0.94,1.24	1.06	0.89,1.27	1.09	0.90,1.32	1.08	0.89,1.31	1.06	0.87,1.29
Relationship with father												
Pretty/very good	0.45	0.28,0.70	0.45	0.28,0.73	0.44	0.23,0.83	0.46	0.23,0.94	0.45	0.24,0.86	0.49	0.25,0.94
Pretty/very bad	0.81	0.36,1.81	0.67	0.28,1.59	1.43	0.53,3.85	1.40	0.47,4.19	0.35	0.08,1.58	0.27	0.06,1.21
Reasonable	Reference		Reference		Reference		Reference		Reference		Reference	
Relationship with mother												
Pretty/very good	0.37	0.23,0.60	0.33	0.20,0.54	0.37	0.19,0.69	0.30	0.15,0.58	0.38	0.18,0.79	0.38	0.18,0.81
Pretty/very bad	1.22	0.54,2.77	0.83	0.33,2.09	0.61	0.13,2.75	0.30	0.04,2.34	1.88	0.66,5.35	1.31	0.43,4.02
Reasonable	Reference		Reference		Reference		Reference		Reference		Reference	

Notes: OR = odds ratios of presenting with physical and mental health conditions, and substance use based on PST use status using non,illicit drug use as a reference; CI = confidence interval; AOR = odds ratios adjusted for age, education, parental status (divorced) and school status (repeating a grade); Bold indicates statistically significant results

Table 4: Associations between physical and mental health, family relationships and substance use and Ecstasy/Amphetamine use in whole sample, boys and girls

	Total n = 101				Ecstasy/Amphetamine users Boys n = 59				Girls n = 52			
	OR	95%CI	AOR	95%CI	OR	95%CI	AOR	95%CI	OR	95%CI	AOR	95%CI
Age	1.08	0.88,1.33			1.13	0.88,1.44			1.00	0.69,1.45		
Education												
Pupil	Reference				Reference				Reference			
Student	1.83	0.74,4.55			1.88	0.57,6;13			1.75	0.42,7.31		
Trainee	1.83	1.04,3.19			2.07	1.09,3.90			0.99	0.24,4.12		
Doubling school	1.55	1.04,2.31			1.52	0.90,2.57			1.53	0.83,2.85		
Parents divorced	1.27	0.82,1.97			1.36	0.77,2.39			1.16	0.58,2.31		
Tobacco Life	9.67	4.49,20.86	9.92	4.32,22.76	8.11	3.24,20.29	8.47	3.04,23.63	13.74	3.32,56.89	12.86	3.09,53.48
Age of first cigarette	0.79	0.72,0.86	0.79	0.72,0.87	0.81	0.73,0.91	0.81	0.72,0.91	0.76	0.66,0.88	0.77	0.66,0.89
Tobacco Daily	4.35	2.94,6.45	4.25	2.76,6.53	4.36	2.60,7.31	4.23	2.37,7.55	4.26	2.32,7.83	4.32	2.26,8.83
Tobacco Intensive	4.22	2.46,7.24	4.18	2.26,7.73	5.47	2.98,10.04	5.47	2.68,11.18	1.61	0.39,6.72	1.85	0.43,7.93
Alcohol Life	12.52	1.75,89.84	11.35	1.57,81.76	6.50	0.90,47.01	5.32	0.73,38.75	NA		NA	
Age of first drink	0.92	0.85,0.99	0.90	0.83,0.96	0.94	0.85,1.03	0.90	0.82,0.99	0.89	0.79,1.00	0.89	0.78,1.00
Alcohol Regular	4.66	2.97,7.31	4.15	2.55,6.77	4.02	2.30,7.02	3.55	1.93,6.53	6.46	2.95,14.17	6.02	2.62,13.86
Alcohol Daily	1.61	0.22,11.72	1.64	0.22,12.05	1.62	0.22,11.85	1.62	0.22,12.05	NA		NA	
Cannabis Life	9.20	5.77,14.67	8.95	5.77,14.67	6.98	3.88,12.57	6.83	3.62,12.87	13.50	6.24,29.22	12.89	5.89,28.23
Age of first joint	0.77	0.67,0.89	0.76	0.65,0.88	0.80	0.67,0.96	0.78	0.64,0.95	0.68	0.51,0.91	0.66	0.49,0.89
Cannabis Regular	4.80	2.54,9.04	4.82	2.46,9.45	4.97	2.49,9.90	5.22	2.50,10.93	2.91	0.39,21.50	3.13	0.42,23.38
Cannabis Daily	2.85	0.89,9.10	2.09	0.50,8.65	3.03	0.94,9.80	2.31	0.55,9.74	NA		NA	
CAST Problematic use	2.16	1.10,4.21	1.98	0.93,4.22	2.49	1.14,5.44	2.10	0.86,5.13	1.79	0.41,7.83	2.26	0.50,10.15
Health self,perception												
Good /very satisfying	Reference		Reference		Reference		Reference		Reference		Reference	
Less/no satisfying	1.06	0.43,2.62	1.17	0.47,2.91	1.27	0.39,4.07	1.45	0.44,4.73	0.88	0.21,3.65	0.92	0.22,3.85
Body shape perception												
Too thin	1.12	0.60,2.08	1.10	0.56,2.18	0.86	0.42,1.78	0.96	0.44,2.09	1.95	0.57,6.65	1.44	0.33,6.30
About the right weight	Reference		Reference		Reference		Reference		Reference		Reference	
Too fat	1.18	0.75,1.85	1.37	0.86,2.19	0.74	0.33,1.66	0.88	0.39,2.00	1.84	0.98,3.46	2.01	1.04,3.87
BMI												
Obese	0.62	0.09,4.47	0.71	0.10,5.16	0.97	0.13,7.09	1.19	0.16,8.76	NA		NA	
Overweight	1.62	0.87,3.01	1.34	0.68,2.63	1.08	0.46,2.55	0.71	0.25,2.01	2.86	1.17,7.01	2.88	1.16,7.16
Normal	Reference		Reference		Reference		Reference		Reference		Reference	
Underweight	1.24	0.71,2.14	1.00	0.54,1.85	1.38	0.64,2.94	1.11	0.47,2.64	1.23	0.55,2.73	0.97	0.39,2.38
Committed suicide lifetime	11.42	6.27,20.80	9.40	4.74,18.66	11.58	4.04,33.17	10.13	3.00,34.17	13.10	6.16,27.88	10.80	4.52,25.81
Age of first suicide	1.03	0.83,1.28	1.06	0.80,1.41	1.14	0.81,1.60	1.23	0.63,2.39	0.97	0.74,1.27	1.02	0.73,1.41
Suicidal ideation past year	2.42	1.45,4.05	2.74	1.62,4.62	1.16	0.42,3.23	1.33	0.48,3.74	4.05	2.12,7.73	4.36	2.24,8.49
See professional mental health	2.21	1.27,3.84	2.30	1.29,4.10	2.17	0.92,5.07	2.58	1.08,6.11	2.44	1.16,5.13	2.21	1.01,4.86

Period of very bad relationship with parents	3.76	2.53,5.60	3.83	2.48,5.93	3.70	2.18,6.28	3.61	2.00,6.51	4.55	2.40,8.61	4.84	2.43,9.62
Age of very bad relationship	1.05	0.93,1.19	1.03	0.90,1.17	1.08	0.90,1.29	1.04	0.86,1.27	1.02	0.86,1.21	1.01	0.84,1.21
Relationship with father												
Pretty/very good	0.43	0.26,0.71	0.44	0.26,0.76	0.65	0.30,1.38	0.93	0.36,2.38	0.27	0.14,0.54	0.23	0.11,0.47
Pretty/very bad	0.97	0.43,2.20	1.06	0.45,2.49	1.43	0.43,4.80	2.13	0.56,8.07	0.71	0.23,2.19	0.71	0.22,2.25
Reasonable	Reference		Reference		Reference		Reference		Reference		Reference	
Relationship with mother												
Pretty/very good	0.44	0.25,0.77	0.47	0.26,0.85	0.44	0.21,0.95	0.45	0.20,1.03	0.43	0.19,0.98	0.48	0.20,1.16
Pretty/very bad	1.25	0.83,4.42	1.94	0.79,4.75	2.74	0.94,8.02	2.47	0.76,8.08	1.21	0.31,4.72	1.43	0.35,5.82
Reasonable	Reference		Reference		Reference		Reference		Reference		Reference	

Notes: OR = odds ratios of presenting with physical and mental health conditions, and substance use based on PST use status using non-illicit drug use as a reference; CI = confidence interval; AOR = odds ratios adjusted for age, education, parental status (divorced) and school status (repeating a grade); Bold indicates statistically significant results